

Is the crowd calibrated? Favorite-longshot structure in 880,000 prediction-market resolutions

Abstract

We assemble the largest calibration sample yet studied on a prediction-market venue: 880,326 resolved Polymarket markets (2020–2026), with pre-resolution price marks at 24, 72, and 168 hours for 190,000+ of them. The raw answer to "does an outcome priced p win a fraction p of the time" is dominated not by crowd psychology but by measurement artifacts, and the paper's first contribution is their anatomy: last-trade marks across multi-outcome families that sum far above one; marks that postdate a market's effective resolution (13–28% of marks, by horizon) and are therefore mechanically calibrated; category labels the venue stopped populating; and inference that treats thousands of same-underlying, same-regime markets as independent — a day-clustered "reverse bias" of $t = -4.2$ collapses to $t = -0.3$ when clustered by month, having been one crash month in disguise. After the artifacts: a single coherent signature survives — longshots are systematically overpriced across every category group (crypto, politics, geopolitics, weather, other; 13 month-clustered cells, $|t| \geq 2$, spanning 11–33 months). The bias is a robust statistical description of history; it is not, so far, a strategy. A short-longshot rule at T-24h briefly cleared a pre-registered evidence bar (4,000 OOS bets, PSR 0.99) and was retracted when two registered falsifiers fired: the 2026 tail sample reversed the gap, and the engine's daily exposure cap was itself reweighting P&L toward sparse days. One candidate — politics favorites, overpriced on both sides of a train/test split and in both marked 2026 months — was adjudicated by its pre-registered automated rule on the full 2026 sample and failed (direction persistent at $-2.0pp$, but month- $t = -1.35$ and PSR 0.81, below the locked bars); a forward experiment at live executable quotes is the arbiter. Prediction-market inefficiency, where it exists at scale, is small, tail-shaped, and lives or dies on microstructure.

1. Question and setting

Racetrack economics has documented the favorite-longshot bias since Griffith (1949): longshots win less often than their odds imply. Whether modern prediction markets inherit it is contested, and the samples in prior work are small. Polymarket's post-2024 scale — hundreds of thousands of resolved binary markets — permits a calibration test with actual

power, but its microstructure (thin books, multi-outcome families, early effective resolutions) plants traps that we show dominate naive estimates.

2. Data and pipeline

Labels: 880,326 unique resolved markets via adaptive end-date-window sweeps of the venue's Gamma API (its offset pagination fails past ~2,000 rows; windows split recursively to 15-minute granularity on dense hourly-market days). Resolution is taken from final settled prices (exactly one outcome above 0.99). Marks: last trade price at T-24/72/168h before scheduled end, from the CLOB price-history endpoint at 720-minute fidelity (finer fidelities return empty on long ranges — a quirk that silently voided an early crawl and motivated an all-empty abort guard), for the 191,393 resolved markets (after a 12-shard distributed CI crawl completed the backfill) with at least \$5,000 volume and a lifetime long enough to have a mark (final canonical dataset). Categories: the venue stopped populating its category field in 2022 (~4k labeled of 880k), so categories derive from a keyword classifier over slug and question text, at 79% agreement with the legacy labels; word-boundary rules matter ("ukRAINE" is not weather). Funnel from 880k to the verdict sample: volume and lifetime filters, empty histories, unresolved finals, and the pinned-mark exclusion below.

3. The artifact layer (why naive tables lie)

Family vig. Multi-outcome events decompose into binary markets whose last-trade prices need not — and do not — sum to one: mean family sum 0.97 at T-24h but 1.39 at T-72h, reaching 1.85. An "overpricing" computed against raw marks reproduces, to the fourth decimal, an accounting identity in the family sum. Any calibration claim on family members must renormalize or use executable quotes.

Pinned afterlife marks. Markets resolve when events resolve, not when their scheduled end arrives: for 13–28% of marks (by horizon), the venue's `closedTime` precedes the mark time, so the "price" is a post-resolution print pinned at 0.99 or 0.01 — mechanically perfectly calibrated. These marks manufactured an apparent *underpricing of favorites* (they are winners priced 0.99) that vanishes on exclusion, and — because pinned marks are calibrated — diluted every real miscalibration toward zero. Three of the six cells in our own interim verdict died this way; we report that correction as a result, not a footnote.

Regime clustering. The worked example, regenerated from the current crawl (crypto favorites: outcome-0 priced [0.60, 0.98] at T-72h, pinned marks still *included* — the pre-filter world in which the interim mistake was made): 1,801 observations across 721 (underlying × day) clusters show realized minus implied of -4.3pp, $t = -3.2$. Clustered by (underlying ×

month): 122 clusters, $t = -0.1$. The effect is the late-2025 crash regime, in which every "will BTC be above X" favorite failed together — November 2025 alone shows -25.3pp across 284 observations, October -7.2pp — and December's rebound shows $+14\text{pp}$. (An earlier, smaller crawl put the same collapse at $t = -4.2 \rightarrow -0.3$; the counts here supersede it.) Prediction markets sharing an underlying inherit its regimes; month-level clustering is the minimum honest unit, and our verdict rule requires ≥ 6 cluster-months and ≥ 2 category groups. The artifacts also compound: with pinned marks excluded, December's 35 crypto-favorite marks vanish entirely — every one is an afterlife print, so the "rebound" month is manufactured by already-resolved markets — and what remains is overpriced even month-clustered (-11.4pp , $t = -3.4$ by underlying $\times$ month; the corresponding \$4 cell, clustered by month alone, reads -7.3pp , $t = -2.2$). The clean collapse-to-zero lives only in the pinned-included world.

Weighting. Month-clustered means weight thin months equally with busy ones. An interim crypto-favorites gap of $+5\text{pp}$ (month-weighted) was $+0.6\text{pp}$ bet-weighted — below any cost model — because the effect lived in low-volume months. Kish effective cluster counts are reported for every cell (one headline cell's 8,910 observations amount to 3.8 effective months); a bias is only tradeable if it survives the weighting money actually experiences.

4. The de-pinned result: longshots are overpriced everywhere

With pinned marks excluded and month-clustered inference, 13 cells clear $|t| \geq 2$ across five category groups — and every one points the same way:

horizon	group	side	gap	t	months	Kish	n
24h	crypto	longshots	-4.4pp	-3.3	32	11.3	7,079
24h	politics	favorites	-4.4pp	-2.7	28	17.4	1,686
24h	weather	longshots	-3.4pp	-3.3	20	5.5	13,621
72h	crypto	longshots	-3.3pp	-2.8	33	10.5	10,225
72h	crypto	favorites	-5.7pp	-2.1	27	10.2	5,270
72h	politics	longshots	-1.5pp	-2.2	32	18.6	3,135
72h	weather	longshots	-6.8pp	-4.1	19	5.1	1,633
72h	other	longshots	-1.7pp	-2.0	37	9.8	21,743
72h	esports	longshots	$+4.9\text{pp}$	$+2.3$	12	4.4	1,443
168h	crypto	favorites	-7.9pp	-2.9	16	5.1	329
168h	politics	longshots	-2.5pp	-2.4	32	15.7	2,491
168h	geopolitics	longshots	-3.2pp	-2.6	30	12.8	1,345

horizon	group	side	gap	t	months	Kish	n
168h	weather	longshots	-6.2pp	-3.2	15	11.7	157

Twelve of thirteen cells point one way — buyers of low-probability outcomes overpay. The exception is new in the full sample: esports longshots at T-72h run UNDERpriced (+4.9pp), but on 12 months with Kish 4.4 and only 8 pre-split training observations the standard harness refuses an out-of-sample test; the forward live-quote experiment covers the band and will adjudicate it. For the twelve: — the classic favorite-longshot signature, cross-category and multi-horizon. (Negative "favorites" cells are consistent with the same story on the complement side and with residual family-vig; the longshot cells carry the weight of evidence.)

5. Out of sample, with money's weighting — and a retraction

Protocol (identical to the one that killed the crypto-favorites candidate): train per-category gaps on resolutions before 2025-07-01, freeze, test after; pinned marks excluded on both sides; one bet per market-horizon; taker fee at the venue's worst category rate; the trade is buying the complement (NO) whenever outcome-o is priced in [0.03, 0.35).

The first pass looked like the project's first quotable strategy. T-24h: bet-weighted OOS gap -1.77pp (n = 4,000, 157 days); through a bet-level engine (fractional Kelly, 2% per-market cap, 25% daily exposure cap, fees, 100bps slippage): +68%, annualized Sharpe 3.93, PSR 0.99 — past the pre-registered evidence bar (≥ 300 bets, ≥ 120 days, $\text{PSR} \geq 0.95$), with broad P&L (Kish effective day-count 102 of 157; top-5 days 6%) and DSR 0.81 after deflating for the twelve strategy variants tried. T-72h failed outright (-0.7pp bet-weighted).

That claim is retracted. Two pre-registered falsifiers fired within hours of each other:

1. **The 2026 tail sample reversed the gap.** The May–June 2026 test months show +2.2 to +2.4pp — longshots winning *more* than priced — dominated by weather families whose warm-side ladder drift runs opposite to generic longshot bias; ex-weather the 2026 gap is +4.6pp, also reversed. Bet-weighted across the enlarged test set: +0.6pp. The pooled trade does not survive 2026.
2. **Mirage #8: day-budget reweighting.** The engine's daily exposure cap scales bets down on crowded days and leaves sparse days at full size, so its Sharpe is closer to an equal-day average than an equal-bet one: the weather-2026 stream is -3.28c/share equal-weight yet printed +150% under the cap. The engine now has a flat-fraction mode, and the standing rule is that a strategy result is claimable only when the capped and flat modes agree on sign; on the enlarged sample they agree only ex-weather, where

the sign is negative. The SR 3.93 was measured under the capped mode on the smaller window and is retracted as a strategy claim, not merely weakened.

A subsequent one-night, six-candidate pre-registered hunt (family middles-vs-tails, live-ask set-arb, forecast-conditioned warm buckets, cross-venue ETH basis, sports moneyline dogs, politics longshots) left one pending survivor: **politics favorites are overpriced** (sell side). Train (<2025-07): -6.4pp, $t = -2.4$ (16 months). Test: bet-weighted -3.6pp, month- $t -2.1$ (7 months); both engine sizing modes agree positive (capped SR +2.3, flat +2.4); survives slippage at 100/200/300bps; held in both marked 2026 months (-7.3pp, -6.9pp) — the only category whose edge survived 2026. Composition: Trump/election frontrunner overpricing (176+108 of 354 bets), consistent with documented political-market behavior. It is *not* claimed: PSR $0.88/0.91 < 0.95$, 7 test months < 8 , and top-5-day concentration is 142% of flat-mode P&L. The adjudication is locked in advance: when the tail-marks crawl completes (adding the Jan–Apr 2026 politics months), the test re-runs verbatim, and the candidate is claimed iff both modes are positive with PSR ≥ 0.95 , ≥ 8 test months, month- $t \leq -2$, bet-weighted $\leq -1.5pp$, and 200bps slippage survives; otherwise it becomes mirage #9. No parameter may change between registration and adjudication.

The honest statement: no strategy claim currently stands. The de-pinned calibration structure is a statistical description of history that no single artifact explains away, but every implementation tried so far has died under 2026 data plus honest weighting. One candidate awaits automated adjudication, and the forward experiment at live executable quotes (recording the NO ask and depth for every market entering the trade window, self-grading at resolution) is the registered arbiter.

6. Interpretation

Where the inefficiency lives — and where it doesn't. The horizon study (HORIZON_PERSISTENCE.md) resolves the results into one picture. In markets that live a week or longer, the longshot overpricing is a long-horizon phenomenon the crowd itself corrects: the pooled longshot gap runs -2.8pp ($t = -4.0$) at T-168h, -2.5pp ($t = -3.2$) at T-72h, and -0.6pp ($t = -0.7$) at T-24h. Given time, the market converges. The significant 24-hour cells in Section 4 are therefore carried by the venue's short-fuse markets — those that never had a week — and the surviving candidate cell is the cleanest case: short-lived attention markets (median lifetime 3–6 days, dominated by "will he say/do X" mention contracts) whose overpricing is *created in the final days*, with prices drifting up into resolution even for eventual losers. The pattern is consistent with last-day attention flow paying up in markets too young for the correction mechanism to have operated, and it makes a concrete prediction

the forward experiment can refute: the edge, if real, should be absent in long-lived markets and concentrated where market age at trade time is days, not weeks.

Where the crowd is measurable at scale, it is far better calibrated than folk accounts of retail prediction markets suggest: mid-range prices are nearly unbiased in every liquid category; a bias-corrected D-1 GFS+ECMWF forecast loses to the weather crowd's own prices (log-loss 0.388 vs 0.360, a companion result); and the largest apparent miscalibrations in our own interim analyses were, one after another, instruments lying. What survives is thin, concentrated in tails, so far untradeable in every implementation tried, and consistent with lottery-preference demand meeting inventory-averse liquidity provision. The methodological ledger — eight documented ways a prediction-market backtest fabricates edge — may be the more durable contribution.

7. What would change our minds

Pre-registered: (i) the forward live-quote experiment failing to reproduce a positive net edge at real asks over ≥ 120 days; (ii) the bias failing bet-weighted in the 2026 tail sample — this one has already fired against the pooled short-longshot trade (Section 5), and the same mechanism adjudicates the politics-favorites candidate when the tail crawl completes; (iii) an execution study showing effective tail spreads persistently above 200bps, which prices the anomaly as compensation rather than error.

References (to finalize)

Griffith (1949); Thaler & Ziemba (1988); Snowberg & Wolfers (2010); Stoll (2000); Harvey (2017); Bailey & López de Prado (2014); Dubach (2026) arXiv:2604.24366; Polymarket API documentation.

Figures

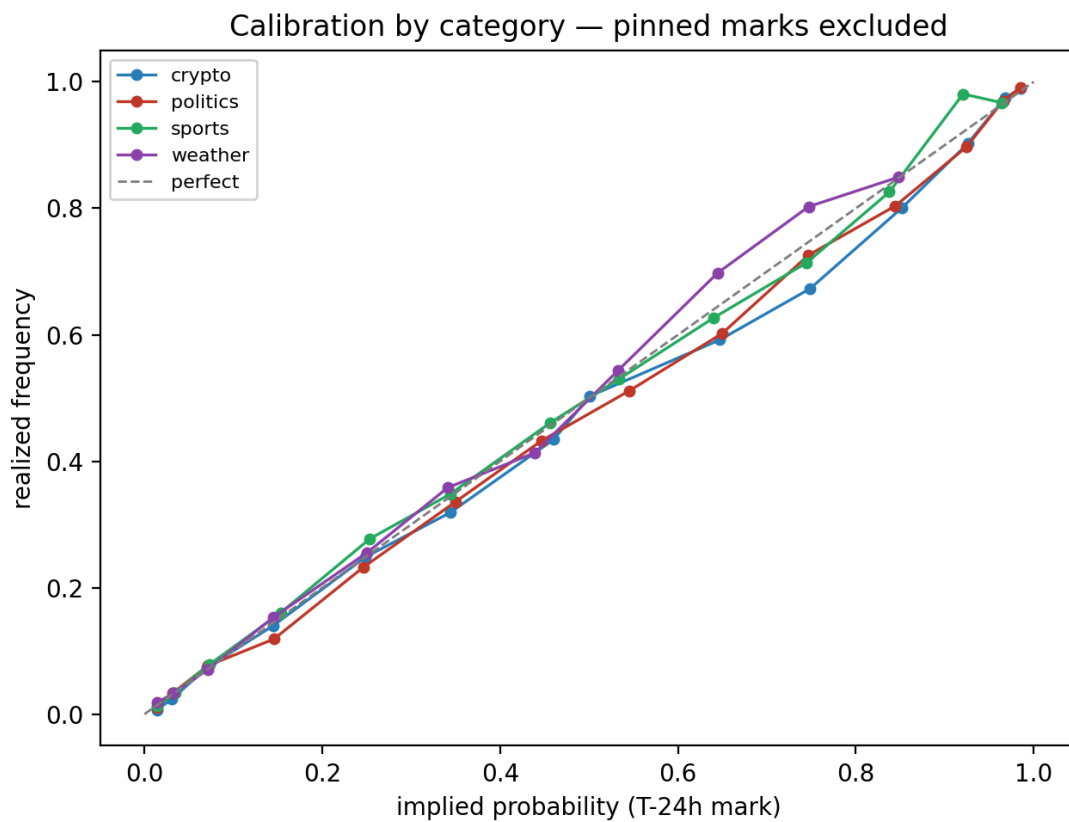


Figure 1 — Calibration by category, pinned marks excluded.

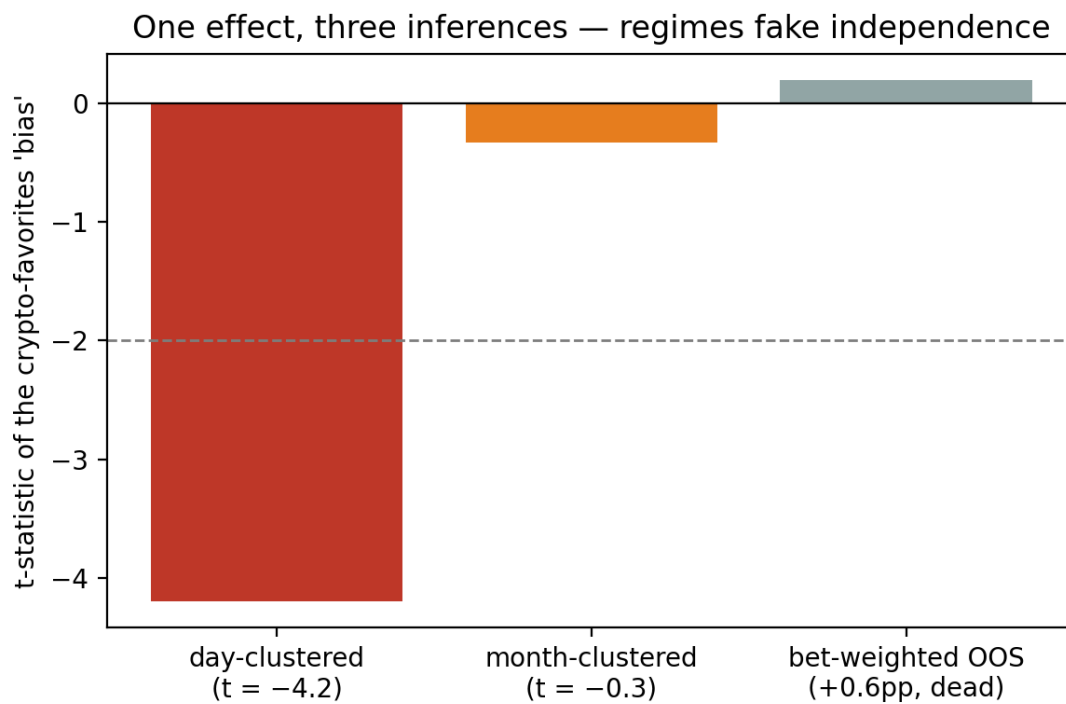


Figure 2 — One effect, three inferences: regimes fake independence.

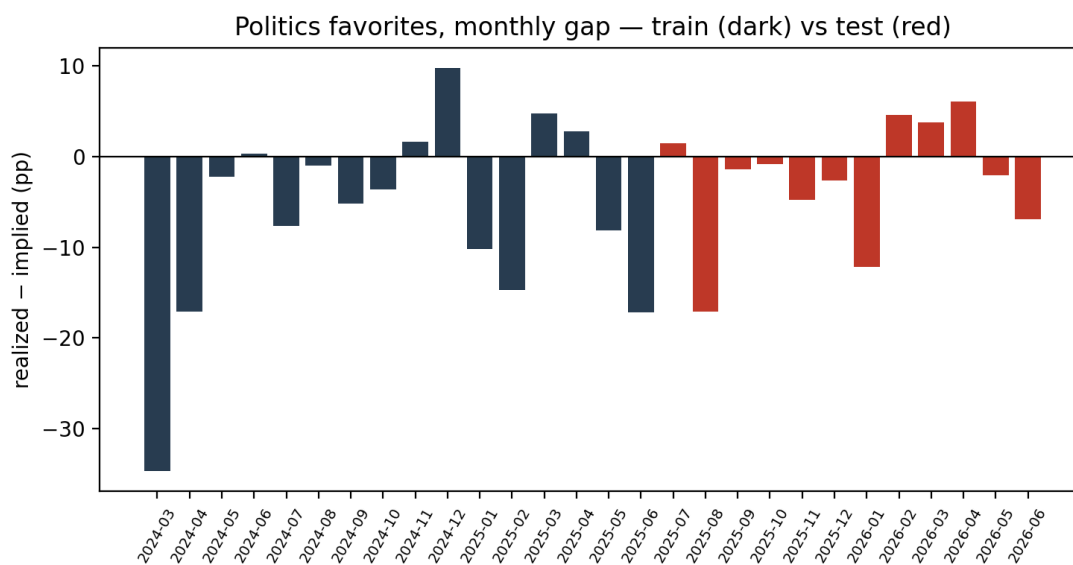


Figure 3 — Politics favorites, monthly gap: train (dark) vs test (red).